

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fifth Semester of B. Tech. (CE) Examination****March-April 2018****CE313/CE302.01/CE302 System Software / System Programming****Date: 31.03.2018, Saturday****Time: 1:30 a.m. To 4:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION – I**Q – 1 Do as directed:**

- (a) Explain the structure of Operating System. **02**
- (b) Differentiate between compiler and interpreter. **03**

- Q - 2 (a)** What is language processor? What are the spectrums of language processor? **04**
- (b) What is input and output of an assembler? Describe the advantages of an assembly language over higher-level language. **04**
- (c) Perform the Lexical, Syntactic and Semantic Analysis on the below given C statement. **06**

```
int A, B;
float C;
B= A + C;
```

OR

- (c) For given grammar, **06**

```
E = T + E | T - E | T
T = T * V | T / V | V
V = a|b|c|d
```

Make Top Down Parsing without backtracking for the string **a + b / c – a * d.**

Q-3 Attempt following questions (any four) **16**

- (a) Prove by an example: Grammar containing left recursion is not enabling to Naïve Top Down Parsing.
- (b) How forward reference problem is solved in one pass assembler? Explain with an example.
- (c) Draw DFA for the string **(0 + 1)* 00**
- (d) Find Register Requirement (RR) for the below given expression

$$f + (x + y) * ((a + b) / (c - d))$$
- (e) What are the attributes and contents of the symbol table? Which data structure is used to implement a symbol table in an efficient way?
- (f) State the responsibilities of the linker in program development. Explain linking process with a block diagram.

SECTION – II

Q - 4 Do as directed:

- (a) What do you mean by program relocation? What do you mean by self-relocating program? **02**
- (b) Differentiate between static memory allocation and dynamic memory allocation. **03**

Q - 5 (a) What the functions are of Pass I assemblers? What are the functions of Pass II assembler? **04**

- (b) What is ambiguous grammar? Consider the following grammar and check whether grammar is ambiguous or not. **04**

$S \rightarrow aSbS \mid bSaS \mid \wedge$

- (c) List the intermediate code generation techniques. Explain any two techniques in brief. **06**

OR

- (c) In block structured language what is the scope of a variable and how to access non local variable? Explain it by example. **06**

Q - 6 Attempt following questions (any four) 16

- (a) What is a loader? What are the functions of loader?
- (b) What is a preprocessor? How does it from macro-processor? Differentiate brief between preprocessor and macro –processor with all the aspects.
- (c) What is a debugger? How debugger helps in the development process?
- (d) What is Editor? What are the functions of Editor?
- (e) Compare call by value and call by reference parameter passing methods.
- (f) Write a short note on nested macro.
